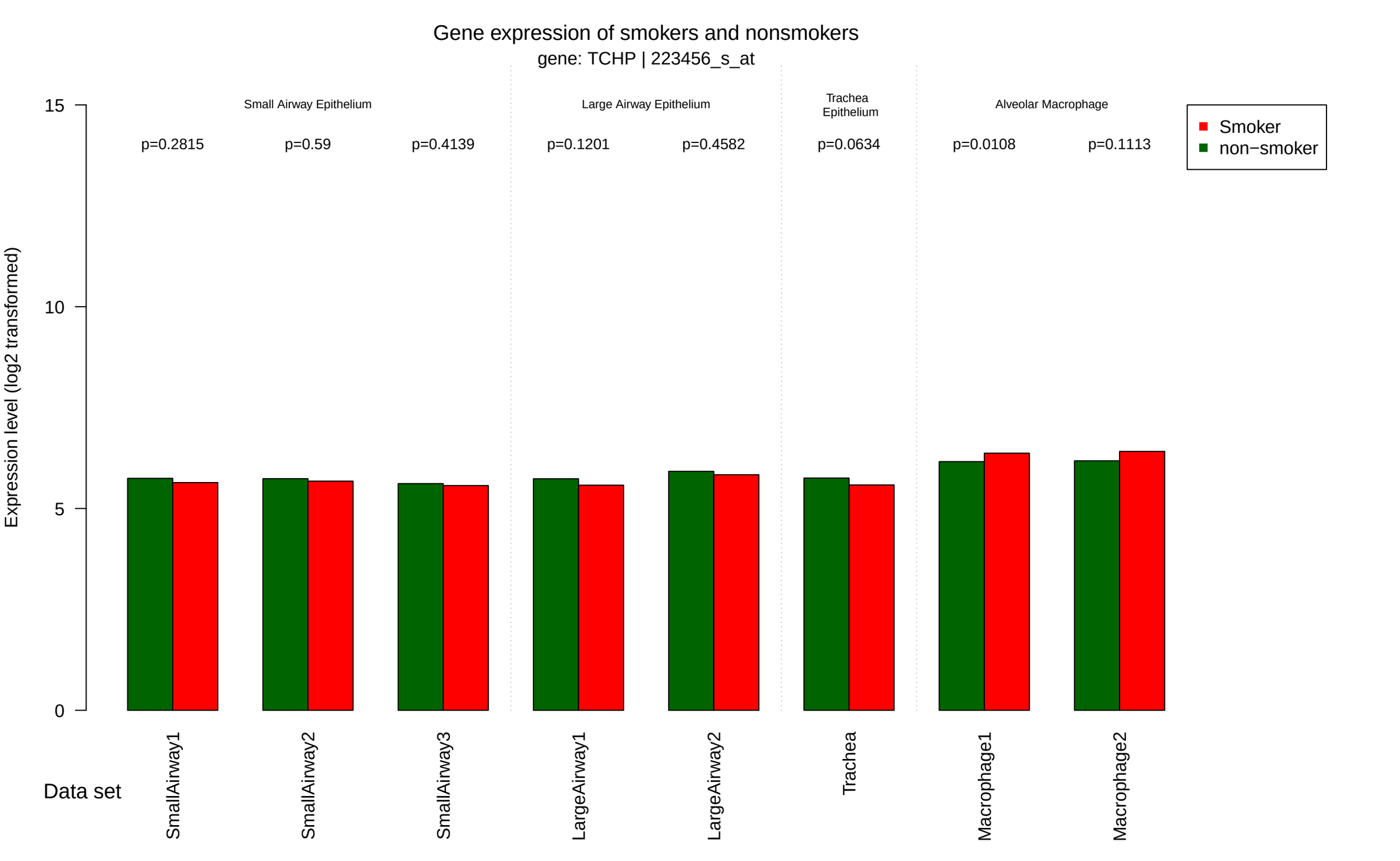
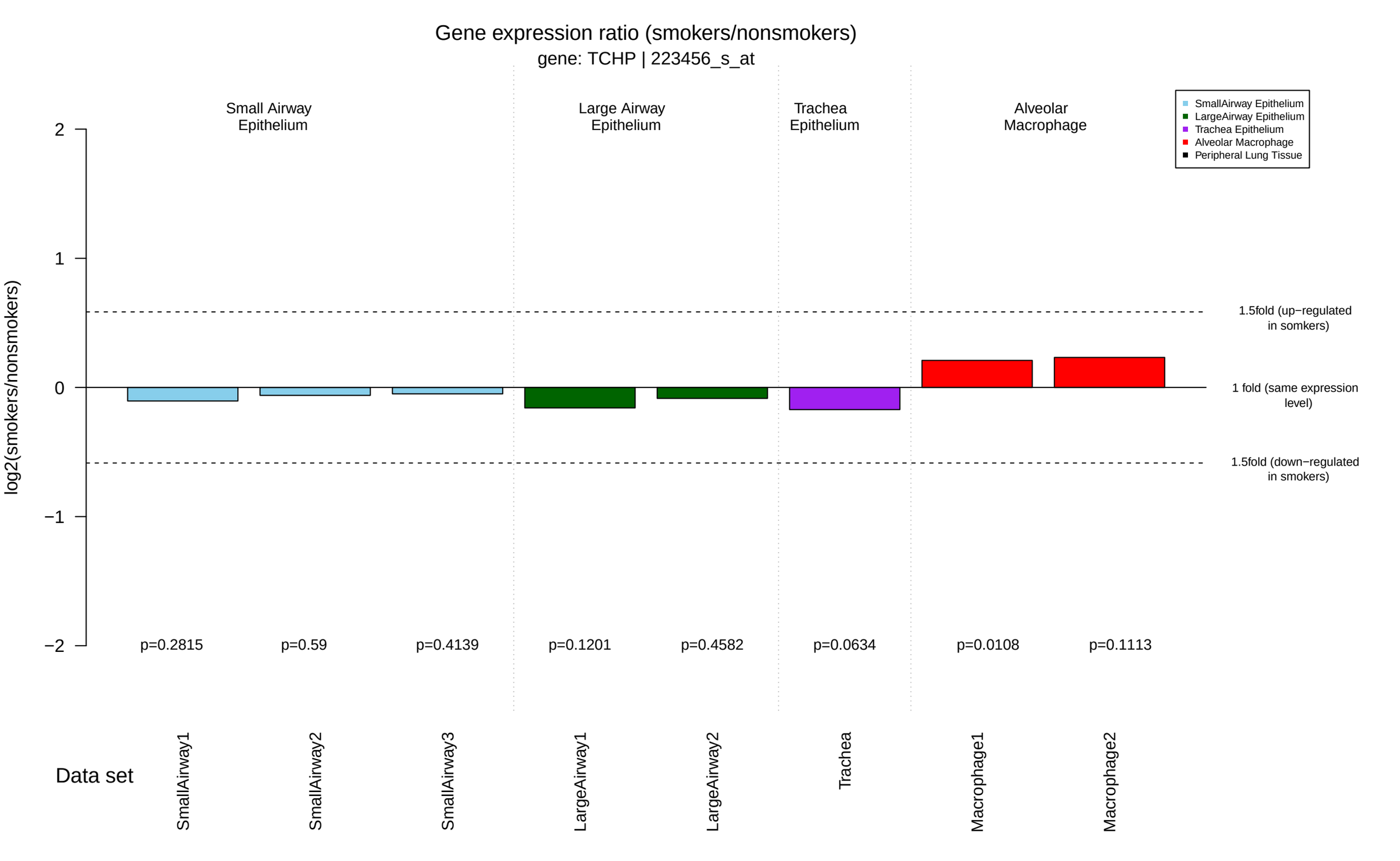
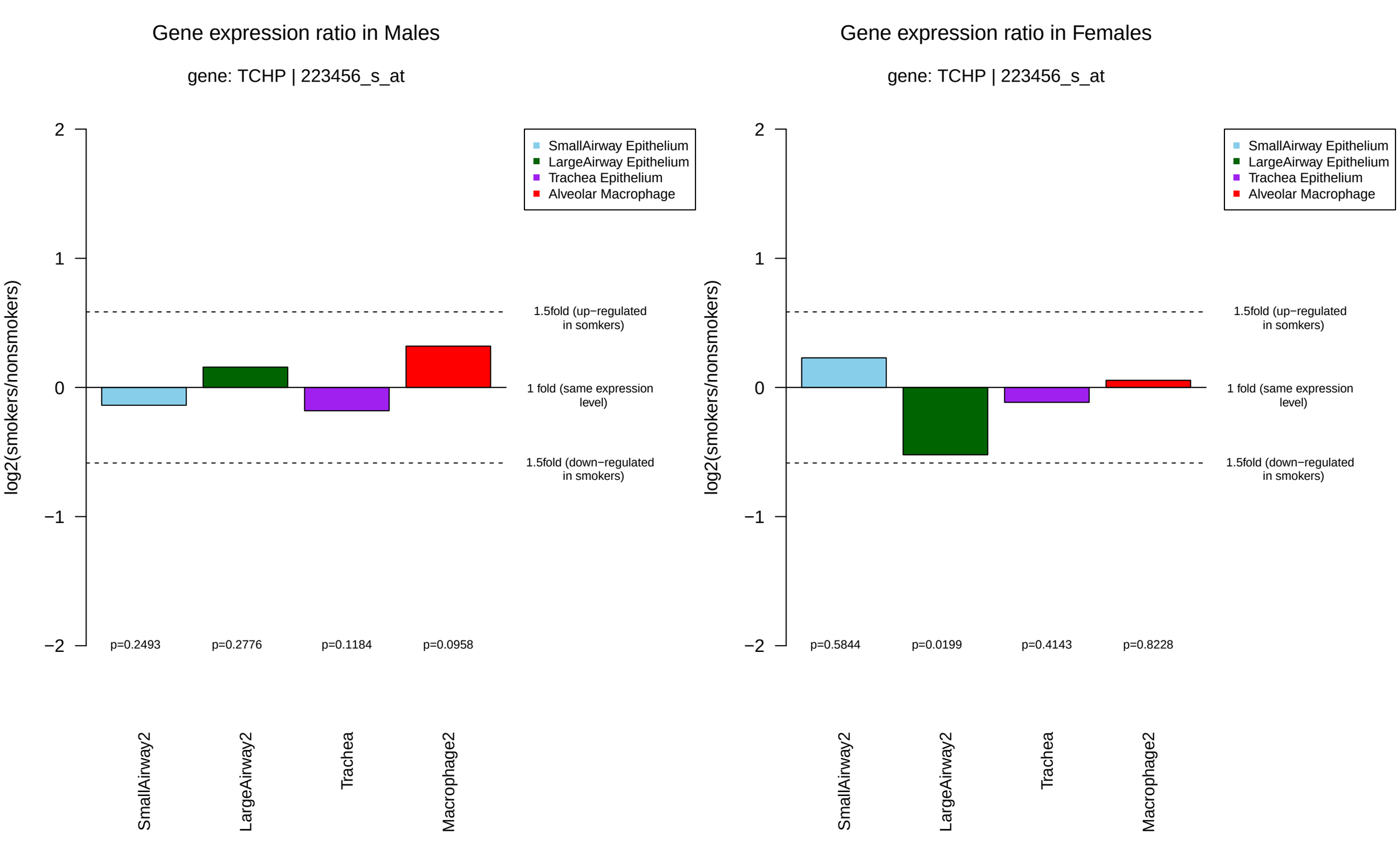
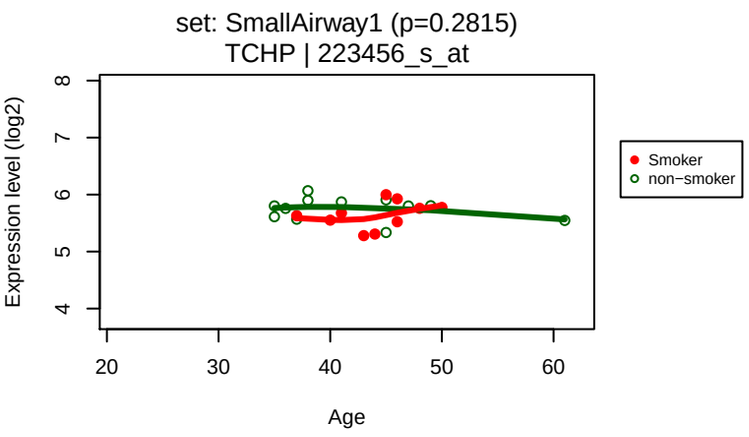
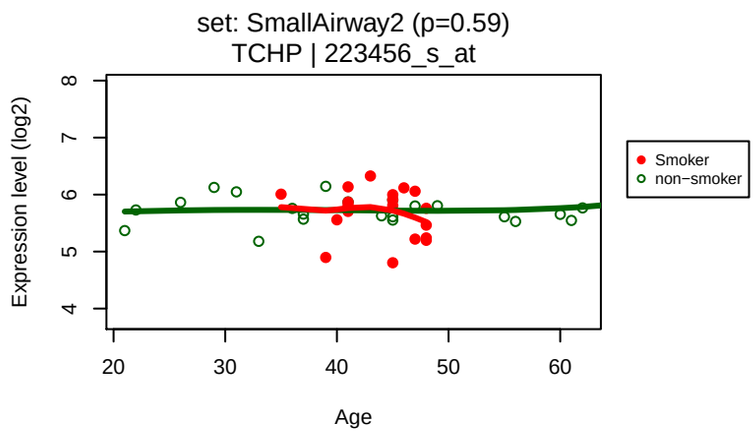
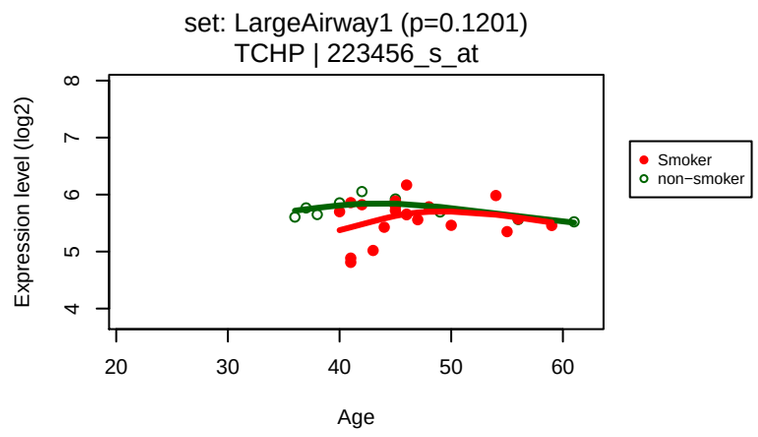

Gene expression of smokers and nonsmokers
gene: TCHP | 223456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2815
p=0.59
p=0.4139
p=0.1201
p=0.4582
p=0.0634
p=0.0108
p=0.1113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

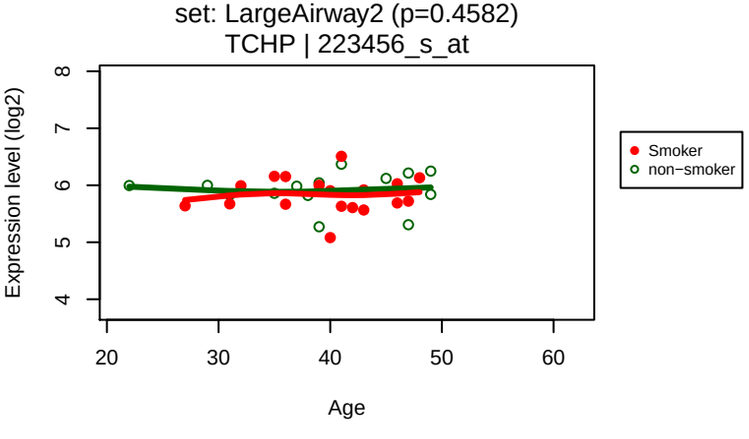

Gene expression ratio (smokers/nonsmokers)
gene: TCHP | 223456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2815
p=0.59
p=0.4139
p=0.1201
p=0.4582
p=0.0634
p=0.0108
p=0.1113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

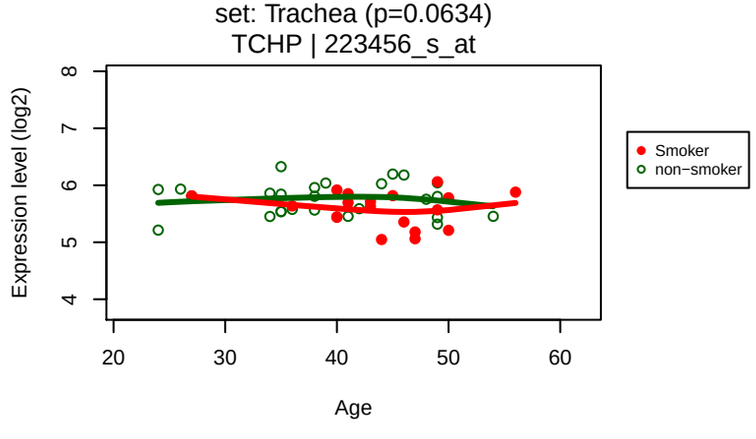

Gene expression ratio in Males
gene: TCHP | 223456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2493
p=0.2776
p=0.1184
p=0.0958
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TCHP | 223456_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5844
p=0.0199
p=0.4143
p=0.8228
SmallAirway2
LargeAirway2
Trachea
Macrophage2

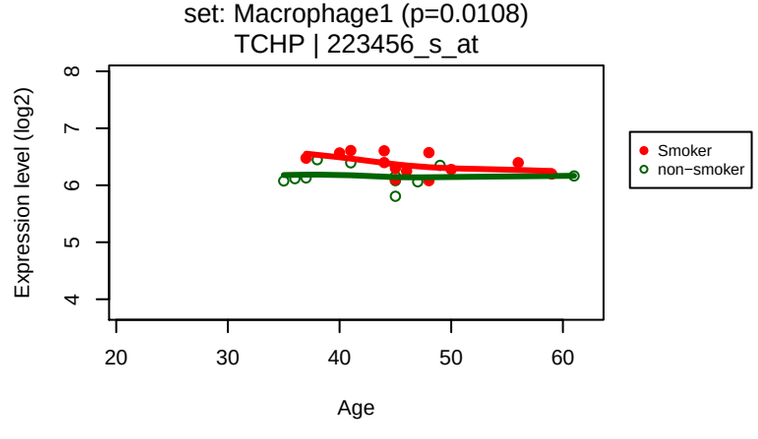

set: SmallAirway1 (p=0.2815)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker

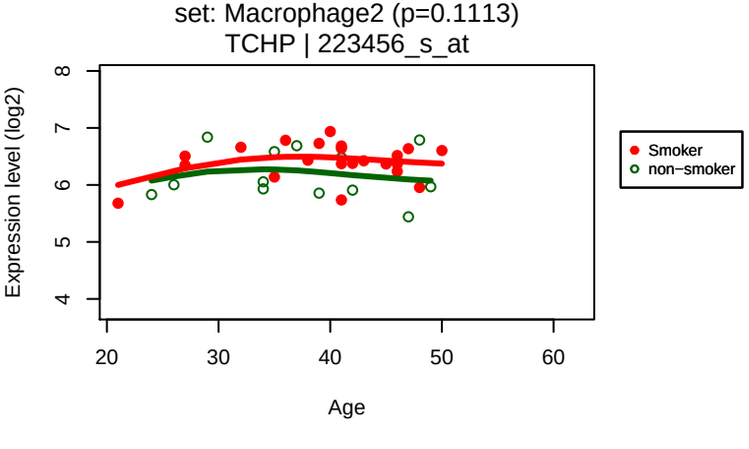

set: SmallAirway2 (p=0.59)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker

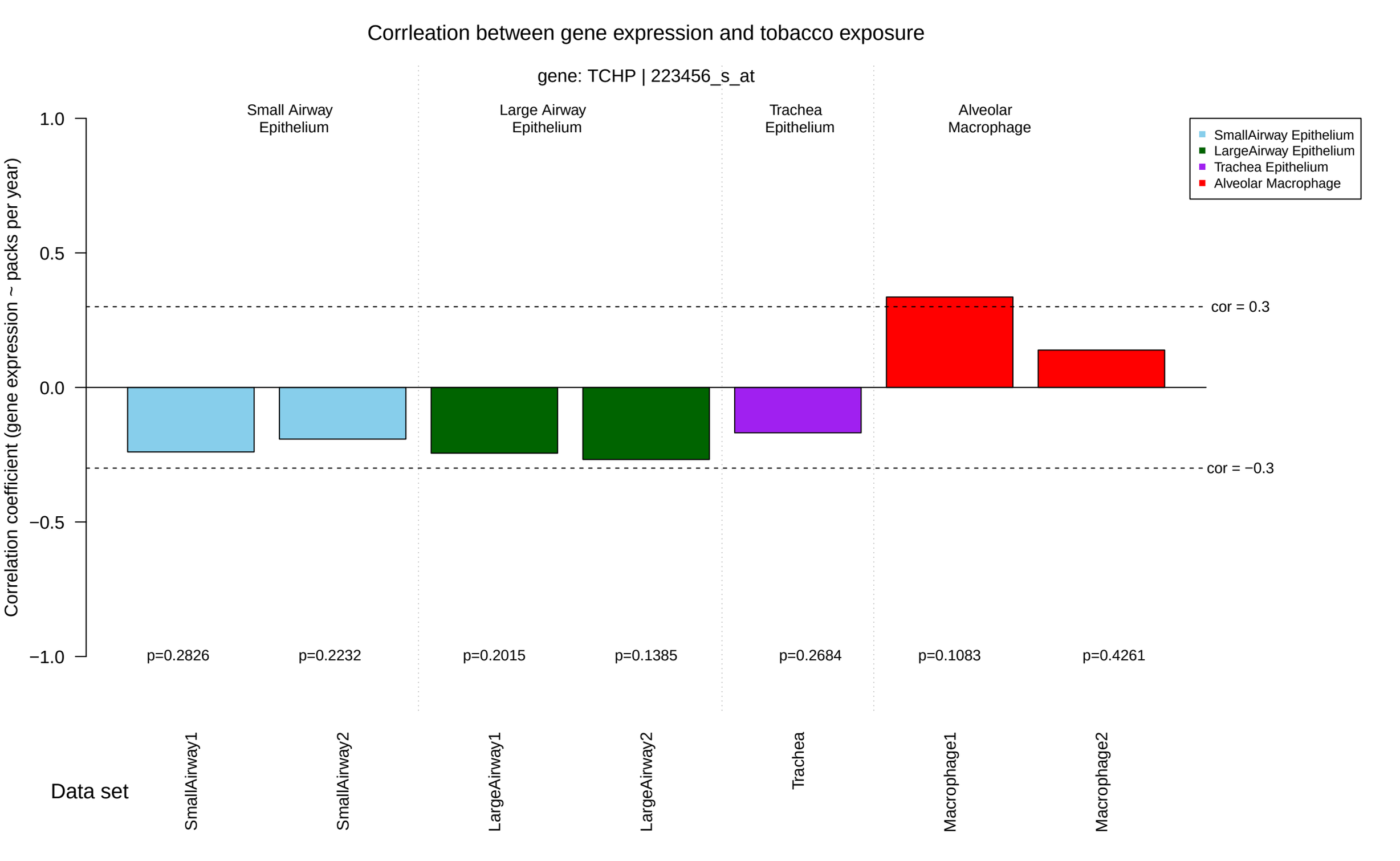

set: LargeAirway1 (p=0.1201)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4582)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0634)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0108)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1113)
TCHP | 223456_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TCHP | 223456_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2826
p=0.2232
p=0.2015
p=0.1385
p=0.2684
p=0.1083
p=0.4261
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2